1. The Hermitian operator is

$$\int \psi^* (\hat{A}\psi)\, d\tau = \int (\hat{A}\psi)^* \psi\, d\tau$$

$$-\frac{\hbar^2}{2m}\int \psi^* \left(\frac{d^2\varphi}{dx^2}\right) dx = -\frac{\hbar^2}{2m}\int \left(\frac{d^2\psi}{dx^2}\right)^* \varphi\, dx$$

To make the problem simplify the T operator reduced to 1-D

$$LHS = -\frac{\hbar^2}{2m}\int \psi^* \frac{d^2\varphi}{dx^2}\, dx = -\frac{\hbar^2}{2m}\left(\left[\psi^* \frac{d\varphi}{dx}\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{d\psi^*}{dx}\frac{d\varphi}{dx}\, dx\right)$$

($\psi^* \to 0$ at $\pm\infty$)

$$= +\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \frac{d\psi^*}{dx}\frac{d\varphi}{dx}\, dx$$

$$= \frac{\hbar^2}{2m}\left(\left[\frac{d\psi^*}{dx}\varphi\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \varphi \frac{d^2\psi^*}{dx^2}\, dx\right)$$

$$= -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \varphi \frac{d^2\psi^*}{dx^2}\, dx = RHS$$

Hence, the kinetic energy operator is a Hermitian Operator.

2) $\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}$ where $m = 0, \pm1, \pm2, \pm3, \pm4, \ldots$

To prove the wavefunction is orthonormal, the following integeral must be satisfied,

$$\int_0^{2\pi} \psi_m^*(\theta)\, \psi_n(\theta)\, d\theta = \delta_{nm} \begin{cases} = 0 & \text{if } m \neq n \\ = 1 & \text{if } m = n \end{cases}$$

for an aurbirary function (θ) the $\psi_m(\theta)$ and $\psi_n(\theta)$ can be

$$\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta} \quad \text{and} \quad \psi_n(\theta) = \frac{1}{\sqrt{2\pi}} e^{in\theta}$$

$$\int_0^{2\pi} \psi_m^*(\theta)\, \psi_n(\theta)\, d\theta = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} e^{-im\theta} \cdot \frac{1}{\sqrt{2\pi}} e^{in\theta}\, d\theta$$

$$= \int_0^{2\pi} \frac{1}{2\pi} e^{i(n-m)\theta}\, d\theta$$

If $m = n$ $= \frac{1}{2\pi} \int_0^{2\pi} d\theta = \frac{1}{2\pi} [2\pi - 0] = 1$

If $m \neq n$ $= \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta}\, d\theta \Rightarrow \frac{1}{2\pi} \int_0^{2\pi} e^{ik\theta}\, d\theta$

k should be an integer $\pm1, \pm2, \ldots$

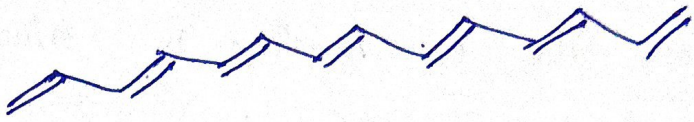
$$= \frac{1}{2\pi} \left[\frac{e^{ik\theta}}{ik} \right]_0^{2\pi} \Rightarrow \frac{1}{2\pi ik} \left(e^{ik2\pi} - e^{0} \right)$$

$$= \frac{1}{2\pi ik} \left(\cos k2\pi + i \sin k2\pi - e^{0} \right) \Rightarrow \frac{1}{2\pi ik} (1-1) = 0$$

Hence, it can be found that if $m = n$ then $\delta_{mn} = 1$ and for $m \neq n$ $\delta_{nm} = 0$.

3. False

4. The conjucated linear polyene containing 7 double bonds is



avarage C–C bond length is 140pm $= 1.4 \times 10^{-10}$ m

14 free electron present in a system since we have 7 double bonds. Each state can hold 2 maximum electrons. The longest transition will be from $n=7$ to $n=8$. $\left(E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{\Delta E}\right)$

Bon length will be $13 \times 140 \times 10^{-10}$ m

$L = 1.82 \times 10^{-9}$ m

$h = 6.626 \times 10^{-34}$ JS
$= 6.626 \times 10^{-34}$ kg m^2 s^{-1}

$m = 9.109 \times 10^{-31}$ kg

$c = 2.99 \times 10^{8}$ ms^{-1}

$$\Delta E = \frac{(n_2^2 - n_1^2)\, h^2}{8 m L^2} \qquad \lambda = \frac{hc\, 8mL^2}{(n_2^2 - n_1^2)\, h^2} \Rightarrow \lambda = \frac{8mL^2 c}{(n_2^2 - n_1^2)\, h}$$

$$\lambda = \frac{8 \times 9.109 \times 10^{-31}\ \text{kg} \times (1.82 \times 10^{-9}\ \text{m})^2 (3 \times 10^{8}\ \text{m s}^{-1})}{(8^2 - 7^2) \times 6.626 \times 10^{-34}\ \text{kg m}^2\ \text{s}^{-1}}$$

$\lambda = 7.28 \times 10^{-7}$ m

$\lambda = 728$ nm

5)

the cyclobutadiene system is [1 1]

So that the length of the square box would be $1.4\text{Å} = 1.4 \times 10^{-10}$ m.

Since, the system contains 4 electrons the HOMO to LUMO transition would be from $n = 2,1 \rightarrow 2,2$

Energy for the transition $\Delta E_{2\rightarrow 3} = \dfrac{(n_2^2 - n_1^2)\, h^2}{8\, m L^2}$

$$\Delta E = \frac{(2^2 - 2^2)\,(6.626\times10^{-34}\ \text{J s})^2}{8 \times 9.109\times10^{-31}\ \text{kg} \times (1.4\times10^{-10}\ \text{m})^2}$$

$$= \frac{5 \times (6.626\times10^{-34})^2\ \text{J}^2\ \text{kg}^2\ \text{m}^2\ \text{s}^{-2}}{(8 \times 9.109\times10^{-31}\ \text{kg})(1.4\times10^{-10})^2\ \text{m}^2}$$

$$\Delta E = E_{2,2} - E_{2,1} = \frac{(8 - 5)\, h^2}{8 m L^2} = \frac{3h^2}{8mL^2}$$

$$= \frac{3 \times (6.626\times10^{-34})^2\ \text{kg}^2\ \text{m}^2\ \text{s}^{-2}}{8 \times 9.109\times10^{-31}\ \text{kg} \times (1.4\times10^{-10})^2\ \text{m}^2}$$

$$= 0.922 \times 10^{-17}\ \text{J} \qquad (1\ \text{eV} = 1.602\times10^{-19}\ \text{J})$$

$$\Delta E = 922\times10^{-18}\ \text{J} = 5.75\ \text{eV}$$

6. True.

As the size of quantum dot increases when we have 10 atoms to 10^5 atoms.

$$\lambda \propto \frac{1}{E} \propto L^2$$

λ increases with length. So, $\lambda_2 > \lambda_1$

7. d. $\theta = 0$ to π and $\phi = 0$ to 2π